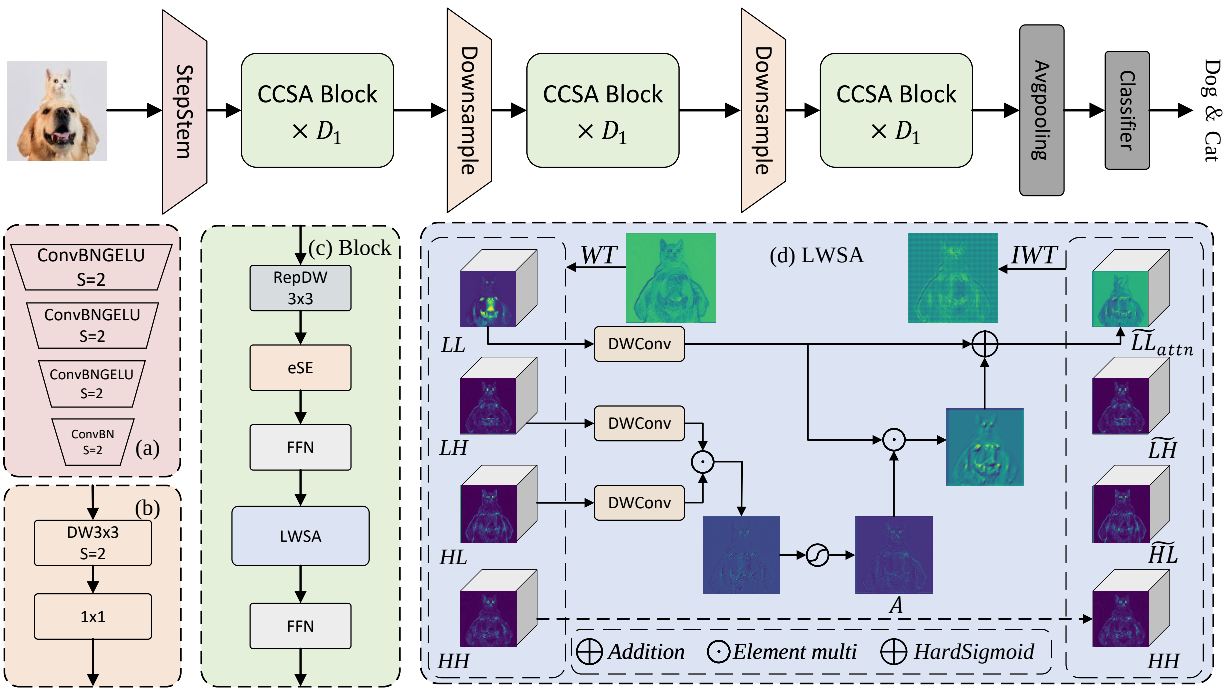
Graphical Abstract

WCANet: Effective Feature Modeling under Low Computational Cost

Fang Wang, Huitao Li, Wenhan Chao, Zheng Zhuo, Xinxin Yang



To design efficient neural networks, many studies focus on reducing redundancy. However, such feature discarding is often random and may remove useful information, which leads to degraded network performance. To address the challenge of insufficient feature extraction under limited computational budgets, we decouple feature modeling into channel-wise and spatial dimensions, which substantially reduces computational overhead. To enable efficient spatial modeling, we propose Learnable Wavelet Spatial Attention (LWSA), which leverages the wavelet transform to decompose the feature map into multiple subbands of varying informativeness and performs spatial attention selectively on the most informative components, enabling effective feature extraction at low computational cost. By cascading the ESE module and LWSA, we propose a Cascaded Channel–Spatial Attention mechanism for feature extraction. Building upon this, we further introduce WCANet, a lightweight architecture that achieves effective feature modeling under low computational budgets across a variety of vision tasks. For example, on ImageNet-1K, WCANet-T5 achieves approximately 42% higher throughput than FasterNet-T1 at comparable accuracy on GPU. WCANet-T4 also achieves 2.9× the throughput of StarNet-S1 with 1.4 percentage points higher Top-1 accuracy, and surpasses MobileMamba-T2 by 1.3 percentage points.

Highlights

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- To overcome the limitation that fixed wavelet filters cannot effectively decompose high-dimensional feature maps, we propose learnable wavelet filters with orthogonality constraints, allowing filters to adapt dynamically while strictly maintaining orthogonality.
- We propose a learnable wavelet spatial attention mechanism (LWSA). LWSA leverages the features from sub-bands with different orientations to perform spatial attention on informative sub-bands, thereby enhancing key spatial nodes and achieving efficient spatial feature modeling.
- We propose Cascade Channel–Spatial Attention (CCSA). CCSA utilizes ESE and LWSA to perceive key features at both channel and spatial levels, respectively, achieving an optimized accuracy-efficiency trade-off.
- We propose WCANet, a visual network design optimized for practical lightweight applications. Extensive experiments demonstrate that WCANet exhibits robust scalability and highly competitive performance across various devices.

WCANet: Effective Feature Modeling under Low Computational Cost

Fang Wang[a,1], Huitao Li[a], Wenhan Chao[b], Zheng Zhuo[a,*] and Xinxin Yang[a]

[a] College of Information Engineering, Beijing Institute of Petrochemical Technology, Beijing, 102617, People's Republic of China
[b] School of Computer Science and Engineering, Beihang University, Beijing, 100191,100083, People's Republic of China

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ABSTRACT

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1. Introduction

With the proliferation of mobile devices, lightweight network designs have become a research hotspot in vision network design [1; 2; 3; 4; 5; 6]. Current mainstream lightweight models are primarily categorized into CNN-based, Transformer-based, and Mamba-based structures. Vision Transformers (ViTs) have drawn significant attention due to their global effective receptive field (ERF) and strong long-range dependency modeling capability. However, the quadratic computational complexity of self-attention can introduce substantial computational overhead as the input resolution increases. Although some works [7; 8; 9; 10; 11; 12; 13] have been proposed to alleviate this issue with notable success, the computational bottleneck of ViTs remains evident under high-resolution inputs.

State-space models (SSMs) [14; 15; 16; 17] are able to capture long-range dependencies with linear computational complexity, motivating researchers to explore their application in vision tasks. Consequently, SSMs have been successfully adapted to the visual domain and have achieved notable results [18; 19]. MobileMamba [20], as a lightweight Mamba-based architecture, demonstrates competitive accuracy and throughput on GPUs. However, practical inference efficiency also depends strongly on hardware platforms and operator support, and the efficiency advantages observed on GPUs may not transfer equally to CPU- or ARM-based devices. Therefore, practical lightweight network design requires consideration of not only theoretical computational complexity but also hardware-aware inference efficiency.

CNN-based methods [21; 11; 22] remain attractive for resource-constrained deployment because of their simple

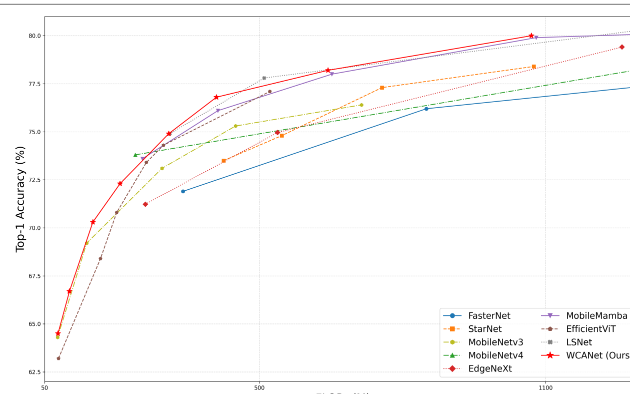

Figure 1: Accuracy–computation comparison between WCANet and representative lightweight vision models on ImageNet-1K. The figure illustrates the accuracy–efficiency trade-off under different computational budgets.

computational structure and broad hardware compatibility. Early lightweight CNNs [23; 24; 25; 26] commonly employed depthwise convolution (DWConv) to reduce computational cost, although lower FLOPs do not necessarily translate into faster practical inference. More recent methods further improve runtime efficiency by reducing spatial computation. For example, FasterNet [27] introduces Partial Convolution (PConv), which applies spatial convolution to only a subset of channels while leaving the remaining channels unchanged. Although such designs effectively reduce computation and memory access, they also reduce the extent of spatial feature extraction, which may limit the modeling of useful spatial information. Therefore, how to reduce computational cost while maintaining sufficient spatial representation capability remains an important challenge in lightweight CNN design.

*Corresponding author
zz@bipt.edu.cn (Z. Zhuo)
ORCID(s): 0000-0001-8028-5285 (Z. Zhuo)

To address the above challenge, we introduce the wavelet transform [28] to improve spatial feature modeling under limited computational budgets by exploiting complementary frequency information. As shown in Fig. 2, the input feature map is decomposed into four sub-bands with different frequency and directional characteristics; additional feature modeling is performed on LL, LH, and HL, while HH is preserved for subsequent reconstruction. The HH sub-band mainly represents diagonal high-frequency components. Rather than discarding this information, LWSA preserves the original HH sub-band and directly routes it to the inverse wavelet transform. Considering computational efficiency, we omit additional convolutional feature extraction on HH, thereby strictly bounding the computational overhead.

Based on the above frequency-aware principles, this paper proposes the Cascade Channel–Spatial Attention (CCSA) module. CCSA decouples feature modeling into channel-wise and spatial dimensions, allowing the two types of information to be modeled separately. Specifically, CCSA first employs the Effective Squeeze and Extraction (ESE) module [29] to enhance informative channels, and then applies Learnable Wavelet Spatial Attention (LWSA) for efficient spatial modeling. LWSA exploits the complementary directional characteristics of the LH and HL sub-bands to generate a spatial attention map, which is applied to the LL sub-band to enhance informative spatial regions. The processed LL, LH, and HL sub-bands are then reconstructed together with the preserved HH sub-band through the inverse wavelet transform.

Based on CCSA, this paper proposes WCANet (Wavelet Cascade Attention Network), a lightweight vision network for resource-constrained scenarios. WCANet is designed to achieve a favorable trade-off between predictive accuracy and practical inference efficiency across different computational budgets. Fig. 1 and Fig. 8 show that WCANet remains competitive with representative SoTA methods in terms of both accuracy–computation and accuracy–inference efficiency trade-offs. For example, on ImageNet-1K, WCANet-T1 achieves 64.5% Top-1 accuracy with only 80M FLOPs, outperforming EfficientViT-M0 by 1.3 percentage points. In lightweight application scenarios with more relaxed computational resources, WCANet also exhibits strong scalability: the Top-1 accuracy of WCANet-T4, T6, and T8 is 3.0, 2.0, and 1.1 percentage points higher than that of FasterNet-T0, T1, and T2, respectively. Moreover, at a comparable accuracy level, WCANet-T5 achieves approximately 42% higher GPU throughput than FasterNet-T1. At comparable GPU throughput, WCANet-T4 achieves 74.9% Top-1 accuracy, outperforming MobileMamba-T2 by 1.3 percentage points, while also achieving higher CPU throughput under the same benchmarking protocol.

Accordingly, the objectives of this work are to improve spatial feature modeling under limited computational budgets, enhance the adaptability of wavelet decomposition to learned feature representations, and achieve a favorable accuracy–efficiency trade-off in practical deployment. The scope of this study focuses on lightweight visual representation learning for resource-constrained scenarios, with evaluations covering image classification, object detection, instance segmentation, medical image segmentation, and inference efficiency on GPU, CPU, and ARM platforms.

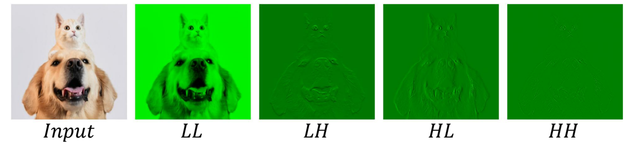

Figure 2: Wavelet decomposition sub-bands. The four sub-bands (LL, LH, HL, HH) are obtained by applying the learnable wavelet transform to an input image.

To summarize, our contributions are as follows:

- To improve the adaptability of wavelet decomposition to learned high-dimensional feature representations, we propose learnable wavelet filters with orthogonality constraints, allowing the filters to adapt during optimization while maintaining an orthogonal wavelet basis.
- We propose Learnable Wavelet Spatial Attention (LWSA), which exploits the complementary directional information of the LH and HL sub-bands to generate spatial attention for the LL sub-band, thereby enhancing informative spatial regions and enabling efficient frequency-aware spatial modeling.
- We propose Cascade Channel–Spatial Attention (CCSA), which combines ESE and LWSA to model informative features along the channel and spatial dimensions, respectively, providing a favorable accuracy–efficiency trade-off.
- Building on CCSA, we develop WCANet, a lightweight visual architecture for resource-constrained scenarios. Extensive experiments across different model scales, vision tasks, and hardware platforms demonstrate its competitive accuracy–efficiency trade-off and practical scalability.

2. Related Work

2.1. Lightweight Network Design

ViT-based methods. Since the Transformer [30] was introduced to computer vision [3], various networks based on the Transformer architecture have demonstrated powerful capabilities in various computer vision tasks [31; 32; 5]. Some works have attempted to combine the ideas of ViT and CNN for application in lightweight tasks [7; 33; 34; 35]. For instance, MobileViT [10] integrates MHSA with MobileNets to achieve an architectural hybrid. EdgeViT [36] proposes the integration of self-attention and convolutions to achieve cost-effective information exchange. In addition, to enhance the actual operational efficiency, MobileViTv2 [33] proposes a linear complexity Transformer architecture to reduce the inference cost. FastViT [34] introduces structural re-parameterization to increase the running speed. SHViT [12] proposes single-head self-attention (SHSA), which improves the operational efficiency by performing attention calculations only on a portion of the channels.

SSM-based methods. State Space Models (SSMs) [14; 16; 17] have attracted considerable interest because of their computational efficiency in modeling long-range dependencies. Mamba [15] incorporates the S6 module, delivering a straightforward architecture that excels in efficiently modeling long sequences. Due to this advantage, many works have applied it to computer vision [37; 38; 18; 39]. Vision Mamba (Vim) [39] adopts bidirectional Mamba block modeling for spatial feature representation. Vmamba [18] introduces a Cross-Scan mechanism to enhance spatial modeling capabilities. EfficientVMamba [19] improves efficiency by using skip sampling. MobileMamba [20] designs a three-stage lightweight architecture with multi-receptive field interaction for high throughput.

CNN-based methods. Since AlexNet [40], CNNs have been widely applied to various computer vision tasks [41; 42; 43; 44; 45; 46]. DWConv [47] has greatly promoted the lightweight development of CNN. MobileNets [23; 21; 11], ShuffleNets [26; 25], and EfficientNet [24] reduce computational redundancy by introducing DWConv and combine ingenious network structure designs to achieve powerful feature extraction capabilities at low computational costs. GhostNet [48] further observes that there is a high degree of redundancy among the channels of feature maps. By performing cheap linear operations only on some channels, it achieves efficient feature extraction capabilities. FasterNet [27] achieves a significant improvement in operational efficiency by using PConv instead of DWConv.

2.2. Wavelet Transforms in Deep Learning

Wavelet transform [28] was introduced to enhance the performance of the model in aspects such as multi-scale feature extraction, computational efficiency, or generation quality. The wavelet transform is used as the pooling operator in convolutional networks [49; 50]. The wavelet transform is also employed to compress feature maps to reduce the computational cost of CNNs [51]. In terms of network architecture design, some researchers have utilized wavelets for downsampling in the U-Net framework [52] and implemented upsampling through inverse wavelet transform [53; 54]. WTConv [55] combines wavelet decomposition with convolution operations, effectively expanding the receptive field and enhancing the response to low-frequency structures. However, most existing methods employ predefined wavelet bases and use wavelet decomposition mainly for pooling, compression, sampling, or convolutional feature extraction. The adaptive use of complementary sub-band information for efficient spatial modeling remains less explored.

2.3. Novelty Compared with Recent Methods

Recent lightweight architectures improve efficiency mainly by reducing the amount or complexity of spatial or global feature modeling. For example, FasterNet [27] applies Partial Convolution to only a subset of channels, SHViT [12] restricts self-attention computation to part of the channels, and MobileMamba [20] adopts lightweight state-space modeling for efficient long-range interaction. In contrast, existing wavelet-based methods such as WTConv [55] primarily employ predefined wavelet decomposition together with convolutional operations for feature extraction and receptive-field enlargement. WCANet differs from these approaches by introducing orthogonality-constrained learnable wavelet filters to adapt frequency decomposition to learned feature representations, while LWSA further exploits the complementary directional information of the LH and HL sub-bands to guide spatial attention on LL. Together with the cascaded channel–spatial modeling in CCSA, WCANet integrates adaptive frequency decomposition and frequency-aware spatial modeling into a unified lightweight architecture under limited computational budgets.

3. Method

3.1. Revisiting Convolutions

The process of the standard convolution operation is as follows: given the input feature map $X \in \mathbb{R}^{C\times H\times W}$ and the Standard Kernel $K \in \mathbb{R}^{C_{out}\times C_{in}\times H\times W}$, the output feature map $Y \in \mathbb{R}^{C_{out}\times H\times W}$ is:

$$Y = X * K, \tag{1}$$

Specifically, $y_{[c',i,j]}$ is an element in the location (i,j) at channel c' of Y. The value of $y_{[c',i,j]}$ is calculated by:

$$y_{[c',i,j]} = \sum_{c=1}^{C_{in}} \sum_{m=1}^{K} \sum_{n=1}^{K} x_{[c,i+m,j+n]} \cdot k_{[c',c,m,n]} \tag{2}$$

As shown in Eq.2, standard convolution performs local spatial feature extraction and channel-wise information fusion in one step [23]. Practice has proved that this way of feature extraction is effective [56; 57]. However, its computation complexity is relatively high $\big(O(HWK^2C_{in}C_{out})\big)$. This makes it difficult to satisfy the lightweight requirements.

The depthwise separable convolution splits the operation of Eq.2 into two layers: a depthwise convolution (DWConv) for filtering and a pointwise convolution (PWConv) layer for combining. The calculations for the two separate layers are respectively shown in Eq.3 and Eq.4.

$$\hat{x}_{c,i,j} = \sum_{m=1}^{K} \sum_{n=1}^{K} x_{c,i+m,j+n} \cdot d_{c,m,n} \tag{3}$$

$$Y = \hat{X} * Conv_{1\times 1}. \tag{4}$$

This decomposition drastically reduces the computation from $(O(HWK^2C_{in}C_{out})$ to $O(HWK^2C_{in}+O(HWC_{in}C_{out})))$.

Partial convolution (Pconv) is proposed to reduce computational redundancy and memory access simultaneously. It simply applies a regular Conv on only a part of the input channels for spatial feature extraction, and leaves the remaining channels untouched.

$$Y = [X_{1:r} * K_{1:r}, X_{r+1:C_{in}}]. \tag{5}$$

The computation complexity of PConv is $O(HWK^2 C_{r\cdot in} C_{r\cdot out})$. Under the same FLOPs, PConv runs faster than DWConv [27].

Although DWConv significantly reduces the computational load, its spatial modeling capability is relatively weak. To make up for the performance loss, in practice, the number of channels of DWConv often needs to be greatly increased, which instead affects the computational efficiency. PConv improves the operational efficiency by only performing convolution operations on a small portion of the feature channels. However, a large amount of potentially useful information in the remaining channels is not extracted, which instead limits the model's feature expression ability.

To achieve truly powerful and efficient feature modeling capabilities under low computational cost, we propose Cascade Channel-Spatial Attention (CCSA). CCSA enables the network to pay more attention to key features when modeling feature maps by cascading channel attention and spatial attention. CCSA reduces computational complexity by separately modeling space and channel. By cascading channel attention and spatial attention, the network pays more attention to key features when modeling feature maps.

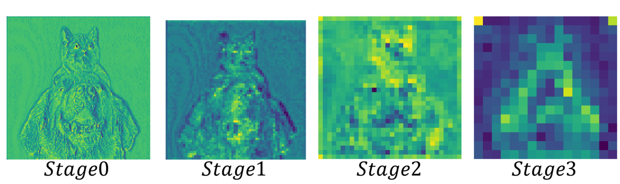

Figure 3: Feature map evolution across network stages. Visualization of feature maps at different stages (Stage 0 to Stage 3).

3.2. Learnable Wavelet Spatial Attention (LWSA)

To achieve a more powerful spatial feature modeling capability, we propose Learnable Wavelet Spatial Attention. By using the wavelet transform [28], the feature map is converted into feature sub-bands containing different information. To enhance computational efficiency, LWSA selectively performs additional feature extraction on the LL, LH, and HL sub-bands, while preserving the HH sub-band through an identity path for subsequent reconstruction. To improve the modeling ability of key spatial features, we take advantage of the characteristic that sub-bands contain different information and construct Wavelet Spatial Attention (WSA).

Previous studies [55; 58; 20] often relied on predefined fixed filters when conducting wavelet transforms, such as Haar Wavelet [51; 59; 60]. After wavelet transformation, the input feature map is transformed into four sub-bands containing different information. Next, existing methods, such as WTConv [55], often extract features from all sub-bands by directly applying standard convolution operations.However, this paradigm introduces two major limitations. First, high-level feature maps contain rich semantic information, as shown in Fig.3. Fixed filters cannot effectively separate the information of high-level feature maps. Second, existing convolution-based feature extraction methods cannot achieve a powerful spatial feature modeling capability. To solve these problems, we innovatively propose learnable wavelet spatial attention (LWSA). It consists of the following two components.

To overcome the first problem, we propose the Learnable Wavelet Basis. Furthermore, we construct **Learnable Wavelet Filters (LWF)** to replace the fixed filters. This not only satisfies orthogonality but also enhances the filter's adaptability. Given an input feature map $X \in \mathbb{R}^{C\times H\times W}$, network generate a two-dimensional wavelet filters group $F \in \mathbb{R}^{4\times2\times2}$ for each channel slice $x \in \mathbb{R}^{1\times H\times W}$. Specifically, we first initialize c one-dimensional learnable low-pass filters $g_{lo} \in \mathbb{R}^2$. The corresponding c high-pass filters g_{hi} are explicitly constructed according to the orthogonality condition:

$$g_{lo}^{\text{init}} = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\end{bmatrix}, g_{hi} = \begin{bmatrix}g_{lo}[1] & -g_{lo}[0]\end{bmatrix}. \tag{6}$$

Building upon this, we construct a two-dimensional analysis filter group F through the matrix multiplication operation:

$$\begin{aligned} &F = \left[F_{\text{LL}} F_{\text{LH}} F_{\text{HL}} F_{\text{HH}}\right] \\ &F_{\text{LL}} = g_{\text{lo}} \otimes g_{\text{lo}}, \quad F_{\text{LH}} = g_{\text{lo}} \otimes g_{\text{hi}}, \\ &F_{\text{HL}} = g_{\text{hi}} \otimes g_{\text{lo}}, \quad F_{\text{HH}} = g_{\text{hi}} \otimes g_{\text{hi}}. \end{aligned} \tag{7}$$

To efficiently implement multi-channel wavelet decomposition, we integrate the c two-dimensional filter groups into a single depthwise separable convolution kernel $W^{DW} \in \mathbb{R}^{4c\times1\times2\times2}$. During the backward pass, g_{lo} is updated by gradient descent. Meanwhile, g_{hi} and W^{DW} are dynamically generated from g_{lo} through orthogonality constraints and tensor operations , which avoids the gradient instability and divergence issues caused by non-orthogonality. A similar hard-constraint design is also reflected in other domains to ensure optimization stability and theoretical validity [61].

To fix the second problem, we propose **Wavelet Spatial Attention** to replace sample concatenation and convolutions. As shown in Fig.4, the LL sub-band retains the most details of the input feature map. The LH sub-band and HL sub-band highlight the vertical and horizontal information, respectively. Therefore, we enhance the information of the LL sub-band by utilizing the information of the LH and HL sub-bands. Fig.5 (c) shows the specific operation. We first perform local convolutional feature extraction on HL and LH by leveraging DWConv, and then perform the Hadamard product [1] on the two obtained feature maps. Finally, the

[1] Here, the symbol $\odot$ strictly denotes element-wise multiplication.

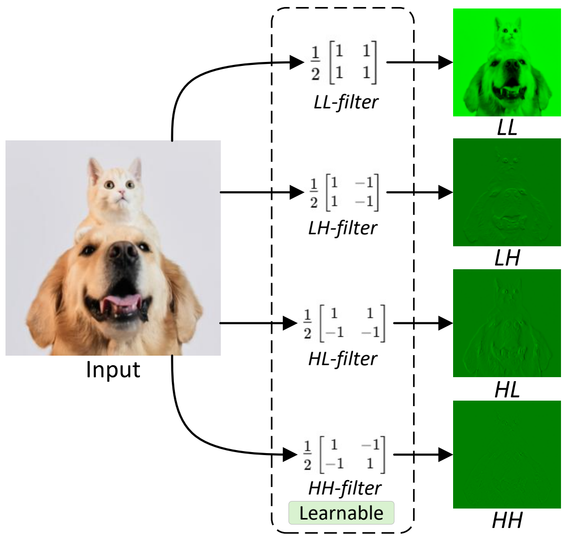

Figure 4: The process of Wavelet Transform.

HardSigmoid activation function is used to generate the spatial attention matrix A.

$$\begin{aligned} \tilde{\mathrm{LH}} &= \mathrm{DWConv}_{3\times3}(\mathrm{LH}) \\ \tilde{\mathrm{HL}} &= \mathrm{DWConv}_{3\times3}(\mathrm{HL}) \\ A &= \mathrm{Hardsigmoid}\left(\tilde{\mathrm{LH}} \odot \tilde{\mathrm{HL}}\right) \end{aligned} \tag{8}$$

where the shape of $A \in \mathbb{R}^{B\times C\times H/2\times W/2}$ is the same as that of the LL sub-band. Thus, spatial attention is achieved through the operation as shown in the formula X:

$$\tilde{\mathrm{LL}}_{\mathrm{attn}} = A \odot \mathrm{DWConv}_{k\times k}(\mathrm{LL}) + \mathrm{LL}, \tag{9}$$

where k represents the size of the convolution kernel. Its value can be dynamically set according to the stage of the network $\left(Stage0 : (k = 7), Stage1 : (k = 5), Stage \geq 2 : (k = 3)\right)$.

$$IWT(\tilde{\mathrm{LL}}_{\mathrm{attn}}, \tilde{\mathrm{LH}}, \tilde{\mathrm{HL}}, HH) = X^{\mathrm{out}} \tag{10}$$

We concatenate $\tilde{\mathrm{LL}}_{\mathrm{attn}}$,$\tilde{\mathrm{LH}}$,$\tilde{\mathrm{HL}}$ with the original HH along the channel-wise, and then perform an inverse wavelet transform to restore the feature map to its original size.

3.3. The design of WCANet

We present the lightweight model design, i.e., WCANet, as shown in Fig.5. It uses a sequence attention as the primary operation. We present the basic block, i.e., CCSA block in Fig.5 (b).

CCSA block applies attention operations separately along the channel and spatial dimensions, which achieves a powerful feature extraction capability. In the channel dimension, the ESE module is employed to suppress redundant channels and enhance effective channels. In the spatial dimension, we employ LWSA to achieve effective spatial feature extraction. Besides, we utilize the extra depthwise convolution and FFN to enhance the model's capability.

Table 1
Comparison of WCANet architectures with different total downsampling ratios: WCANet-T4×32 and WCANet-T5x32 denote 4-stage networks with a total downsampling factor of 32, while WCANet-T4×64 and WCANet-T5×64 denote 3-stage networks with a total downsampling factor of 64.

model	network structure	Throughput (images/s)	FLOPs (M)
WCANet-T4 ×32	{32, 64, 128, 256} {1, 2, 4, 5}	5334	275
WCANet-T4 ×64	{128, 256, 512} {2, 3, 2}	9921	313
WCANet-T5 ×32	{40, 80, 160, 320} {1, 2, 4, 5}	4112	442
WCANet-T5 ×64	{128, 256, 512} {3, 4, 3}	7472	410

We compare the ×32 and ×64 downsampling multi-stage architectures. As shown in Tab 1, under different network architectures, ×64 downsampling is more efficient than ×32 downsampling. Furthermore, the experimental results show that WCANet-T4×64 achieves a Top-1 accuracy 0.8 ↑ than its ×32 counterpart, and WCANet-T5×64 matches the Top-1 accuracy of the ×32 version. Therefore, we utilize the 64× downsampling in WCANet.

Overall, WCANet employs a StepStem [62] to project the input image into the visual feature map. For downsampling, we leverage the depth-wise and point-wise convolution to reduce the spatial resolution and modulate the channel dimension, respectively. Besides, we stack CCSA blocks in all stages. Finally, the feature map is processed by global average pooling and a classifier to produce the final class prediction.

We build multiple WCANet variants to accommodate different computational budgets. Under an input resolution of 256×256, their FLOPs range from 80M to 650M, namely WCANet-T1, WCANet-T2, WCANet-T2d5, WCANet-T3, WCANet-T4, WCANet-T5, and WCANet-T6. Additionally, to better suit downstream dense prediction tasks, we also design a four-stage WCANet variant with 64× downsampling: WCANet-T8. Detailed stage-wise architectural configurations, including feature resolutions, channel dimensions, and numbers of CCSA blocks, are provided in Tables 13 and 14 in the Appendix.

4. Experiments

To comprehensively evaluate WCANet under different visual tasks and computational conditions, we conduct experiments on ImageNet-1K for image classification, COCO-2017 for object detection and instance segmentation, and ClinicDB for medical image segmentation. We additionally evaluate hardware-aware inference efficiency on GPU, x86 CPU, and ARM-based platforms. For the relatively small

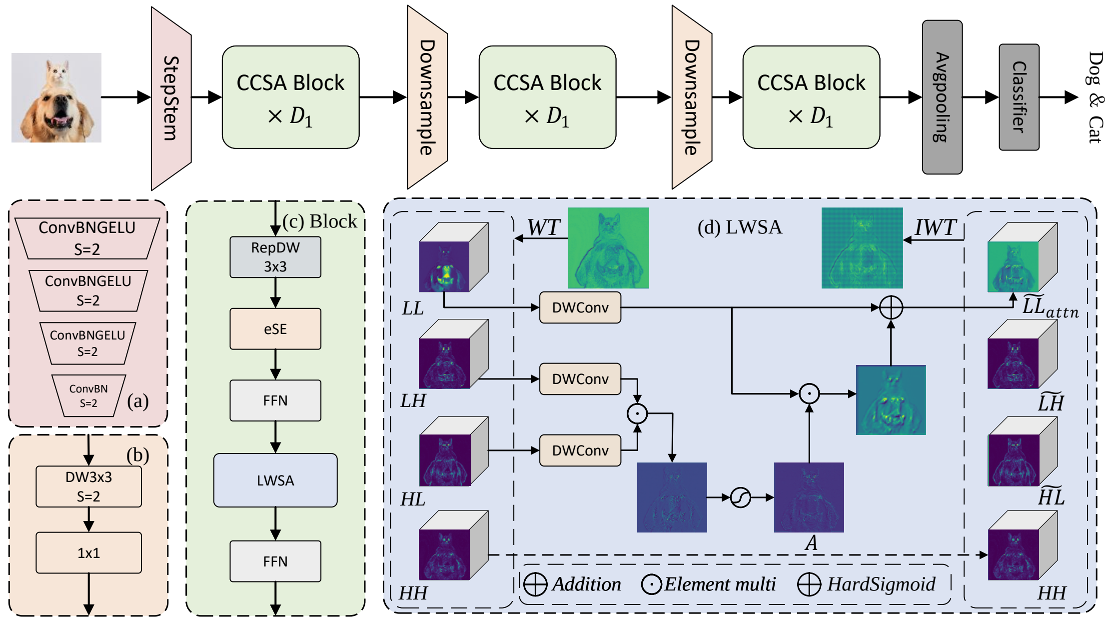

Figure 5: The structure of WCANet. (a) is the StepStem. (b) is the CCSA Block. (c) is the LWSA.

ClinicDB dataset, robustness to dataset partitioning is further assessed over five independently generated data splits.

We adopt task-specific standard evaluation metrics. Top-1 accuracy is used for ImageNet-1K classification, together with model parameters, FLOPs, and measured throughput for computational and practical efficiency evaluation. For COCO-2017, we report standard AP metrics at different IoU thresholds and object scales for both bounding-box detection and instance segmentation. For ClinicDB, Dice and IoU are used to evaluate the overlap between the predicted and ground-truth segmentation masks.

4.1. Image Classification

We conduct experiments on the ImageNet-1K [68] dataset to evaluate the performance of WCANet on the image classification task. ImageNet-1K contains 1,281,167 training images and 50,000 validation images from 1,000 object categories. The number of training images per category ranges from 732 to 1,300, while the validation set contains 50 images for each category. We follow the standard ImageNet-1K training and validation splits.

To ensure a fair and rigorous comparison against baselines spanning different publication years, we explicitly document our training protocol. Specifically, WCANet is trained entirely from scratch without relying on Knowledge Distillation (KD), additional pre-training datasets, or multi-scale training procedures. The detailed hyperparameters and data augmentation strategies are fully documented in Table 12. The overall training protocol is similar to that of [9; 66; 20], except that WCANet uses an input image resolution of 256×256 and a learning rate that is cosine-annealed from 3e-3 to 1e-5. As the stage-wise output stride of WCANet reaches $\times 64$, the deepest-stage feature map has a spatial resolution of $H/64 \times W/64$. Within the LWSA block, the wavelet decomposition further reduces the spatial resolution by a factor of 2, resulting in a minimum internal resolution corresponding to an effective downsampling factor of $\times 128$. Therefore, we adopt an input resolution of 256×256, which is divisible by 128, to maintain exact spatial alignment during wavelet decomposition and inverse reconstruction.

Table 2 shows the performance of WCANet, compared with other state-of-the-art (SoTA) methods under similar model scales. The different model scales are categorized by FLOPs. Compared with CNNs, WCANet demonstrates consistently superior performance at the same computational cost. Specifically, WCANet-T4 outperforms FasterNet-T0 [27] by +3 ↑. WCANet-T8 surpasses FasterNet-T2 [27] by 1.1 ↑ in Top-1 while having only 56% of its FLOPs. Furthermore, WCANet variants consistently surpass StarNet [65] counterparts by 1.2 ↑ to 3.3 ↑ in Top-1. Compared with ViTs, ViMs, and hybrid architectures, WCANet also demonstrates strong competitiveness. Under comparable FLOPs,

WCANet variants consistently outperform the Transformer-based EfficientViT [9] series in Top-1 accuracy. WCANet-T4 achieves 74.9% Top-1 accuracy, surpassing the Mamba-based MobileMamba-T2 [20] by +1.3 ↑. At approximately 1 GFLOPs, WCANet-T8 achieves 80.0% Top-1 accuracy, outperforming contemporary architectures at similar computational costs, including MobileMamba-B1 [20], EMO-6M [64], and ShViT-S4 [12]. These experimental results strongly indicate that by employing LWSA for spatial feature extraction and the ESE module for inter-channel information

Table 2
Classification results of different models on ImageNet-1K.

Model	Params (M)	FLOPs (M)	Reso.	Top-1 (%)	#Pub
EfficientViT-M0 [9]	2.3	79	224	63.2	CVPR'23
MobileNetv3-Lx035 [63]	2.1	77	224	64.2	ICCV'19
WCANet-T1	2.0	78	256	64.5	-
WCANet-T2	2.5	102	256	**66.7**	-
EfficientViT-M1 [9]	3.0	167	224	68.4	CVPR'23
MobileNetv3Lx05 [63]	2.6	138	224	68.8	ICCV'19
ShufflenetV2-1.0x [25]	2.3	146	224	69.4	ECCV'18
EfficientViT-M2 [9]	4.2	201	224	70.8	CVPR'23
WCANet-T2d5	3.7	151	256	70.3	-
WCANet-T3	5.1	208	256	**72.3**	-
EdgeNeXt-XXS [8]	1.3	260	224	71.2	ECCV'22
FasterNet-T0 [27]	3.9	340	224	71.9	CVPR'23
ShuffleNetV2-1.5x [25]	3.5	300	224	72.6	ECCV'18
EMO-1M [64]	1.3	261	224	71.5	ICCV'23
SHViT-S1 [12]	6.3	241	224	72.8	CVPR'24
EfficientViT-M3 [9]	6.9	263	224	73.4	CVPR'23
StarNet-S1 [65]	2.9	425	224	73.5	CVPR'24
MobileMamba-T2 [20]	8.8	255	192	73.6	CVPR'25
EfficientViT-M4 [9]	8.8	299	224	74.3	CVPR'23
LSNet-T [66]	11.4	320	224	74.9	CVPR'25
EMO-2M [64]	2.3	439	224	75.1	ICCV'23
SHViT-S2 [12]	11.4	366	224	75.2	CVPR'24
MobileMamba-T4 [20]	14.2	413	192	76.1	CVPR'25
WCANet-T4	7.6	313	256	74.9	-
WCANet-T5	10.8	410	256	**76.8**	-
StarNet-S2 [65]	3.7	547	224	74.7	CVPR'24
EdgeNeXt-XS [8]	2.3	540	224	75.0	ECCV'22
FasterNet-T1 [27]	7.6	850	224	76.2	CVPR'23
EfficientViT-M5 [9]	12.4	522	224	77.1	CVPR'23
SHViT-S3 [12]	14.2	601	224	77.4	CVPR'24
MobileMamba-S6 [20]	15.0	652	224	78.0	CVPR'25
WCANet-T6	16.5	644	256	**78.2**	-
StarNet-S3 [65]	5.8	757	224	77.4	CVPR'24
RepViT-M0.9 [67]	5.1	846	224	77.4	CVPR'24
EdgeViT-XS [36]	6.7	1100	224	77.5	ICCV'22
StarNet-S4 [65]	7.5	1075	224	78.8	CVPR'24
FasterNet-T2 [27]	15.0	1917	224	78.9	CVPR'23
EMO-6M [64]	6.1	961	224	79.0	ICCV'23
SHViT-S4 [12]	16.5	986	256	79.4	CVPR'24
EfficientViT-M4 [9]	12.4	1486	384	79.8	CVPR'23
MobileMamba-B1 [20]	17.1	1080	256	79.9	CVPR'25
WCANet-T8	26.0	1078	256	**80.0**	-

fusion, the proposed model performs very well in terms of classification accuracy.

4.2. Inference Efficiency Analysis

To evaluate the hardware-aware inference efficiency of WCANet, we conduct throughput benchmarks on three representative hardware platforms: an NVIDIA GeForce RTX 4090 GPU, an Intel(R) Xeon(R) Platinum 8352V CPU, and an Apple M1 (ARM-based chip). The batch sizes are set to 2048, 16, and 8 for the GPU, CPU, and Apple M1 platforms, respectively. Each model is evaluated at the input resolution listed in Table 3. For each model-device pair, the throughput measurement is repeated five times, and the results are reported as mean ± standard deviation. All models are benchmarked in PyTorch using FP32 precision, and all operators are implemented with official library components without additional model-specific deployment optimizations.

Table 3
Comparison with SoTAs in inference efficiency. GPU throughput is measured on an NVIDIA GeForce RTX 4090 with a batch size of 2048; CPU throughput is evaluated on an Intel(R) Xeon(R) Platinum 8352V with a batch size of 16; and ARM throughput is evaluated on an Apple M1 chip with a batch size of 8. Each model is evaluated at the input resolution listed in the "Reso." column. Throughput is reported as mean ± standard deviation over five repeated runs.

Model	FLOPs (M)	#Params (M)	Reso.	Throughput (img/s) GPU	CPU	ARM	Top-1 %
StarNet-S1	425	2.9	224	6898±55	42.7±0.8	25.8±0.5	73.5
FasterNet-T0	340	3.9	224	15682±141	77.4±1.2	62.6±1.0	71.9
MobileMamba-T2	255	8.8	192	19541±127	54.6±1.0	51.7±0.9	73.6
LSNet-T	320	11.4	224	16106±134	-	-	74.9
WCANet-T4	313	7.6	256	20274±146	60.6±1.1	48.7±0.8	74.9
FasterNet-T1	850	7.6	224	10731±163	38.7±0.7	42.6±0.8	76.2
MobileMamba-T4	413	14.2	192	15418±128	40.7±0.8	38.8±0.7	76.1
WCANet-T5	410	10.8	256	15241±143	41.7±1.4	34.8±0.6	76.8
FasterNet-T2	1917	15.0	224	4946±95	18.8±0.8	27.7±0.6	78.9
StarNet-S4	1075	7.5	224	3492±43	23.8±0.5	11.9±0.4	78.8
MobileMamba-B1	1080	17.1	256	6152±52	12.8±0.4	15.8±0.7	79.9
WCANet-T8	1078	26.0	256	6236±57	17.8±0.4	16.8±0.3	80.0

As summarized in Table 3, WCANet achieves a favorable accuracy–efficiency trade-off across different hardware platforms. On the RTX 4090, WCANet-T5 achieves a throughput of 15,241 ± 143 img/s with 76.8% Top-1 accuracy, compared with 10,731±163 img/s and 76.2% Top-1 accuracy for FasterNet-T1. Thus, at a comparable accuracy level, WCANet-T5 provides approximately 42% higher throughput. In addition, WCANet-T4 achieves 20,274 ± 146 img/s, which is approximately 2.9× the throughput of StarNet-S1 (6,898 ± 55 img/s), while improving Top-1 accuracy by 1.4 percentage points.

The performance trade-offs are further visualized in Figure 8, which plots the relationship between inference latency/throughput and Top-1 accuracy across different hardware platforms. As illustrated, WCANet consistently lies near the favorable frontier of the accuracy–efficiency trade-off compared with FasterNet, StarNet, and MobileMamba. These results indicate that WCANet effectively balances feature modeling capability and computational overhead, making it suitable for resource-constrained applications where both predictive accuracy and hardware-specific inference efficiency are important.

4.3. Object Detection and Segmentation on COCO

To further evaluate the transferability of WCANet to dense prediction tasks, we conduct object detection and instance segmentation experiments on the COCO-2017 dataset [71]. We use the standard COCO-2017 training and validation splits, containing 118,287 and 5,000 images, respectively. COCO contains 80 object categories with bounding-box and instance-level segmentation annotations,

Table 4
Object detection and instance segmentation results on COCO. AP^b and AP^m indicate bounding box AP and mask AP, respectively. For a fair comparison, all baselines are evaluated according to the training protocols reported in their original publications.

Backbone	RetinaNet						Mask R-CNN					
	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AP^b	AP^b_{50}	AP^b_{75}	AP^m	AP^m_{50}	AP^m_{75}
EfficientViT-M5 [9]	34.3	54.2	36.1	18.0	36.9	48.2	34.9	57.0	37.0	32.8	53.7	34.6
SHViT-S3 [12]	36.1	56.6	38.0	19.9	39.1	50.8	36.9	59.4	39.6	34.4	56.3	36.1
ResNet50 [2]	36.3	55.3	38.6	19.3	40.0	48.8	38.0	58.6	41.4	34.4	55.1	36.7
PVT-Tiny [69]	36.7	56.9	38.9	**22.6**	38.8	50.0	36.7	59.2	39.3	35.1	56.7	37.3
PoolFormer-S12 [70]	36.2	56.2	38.2	20.8	39.1	48.0	37.3	59.0	40.1	34.6	55.8	36.9
LSNet-S [66]	36.7	67.2	38.6	20.0	39.7	51.8	37.4	59.9	39.8	34.8	56.8	36.6
FasterNet-S [27]	-	-	-	-	-	-	39.9	61.2	**43.6**	**36.9**	58.1	**39.7**
WCANet-T8	**38.6**	**59.0**	**41.1**	22.5	**41.5**	**51.7**	**40.0**	**61.3**	43.4	36.7	**58.6**	39.1

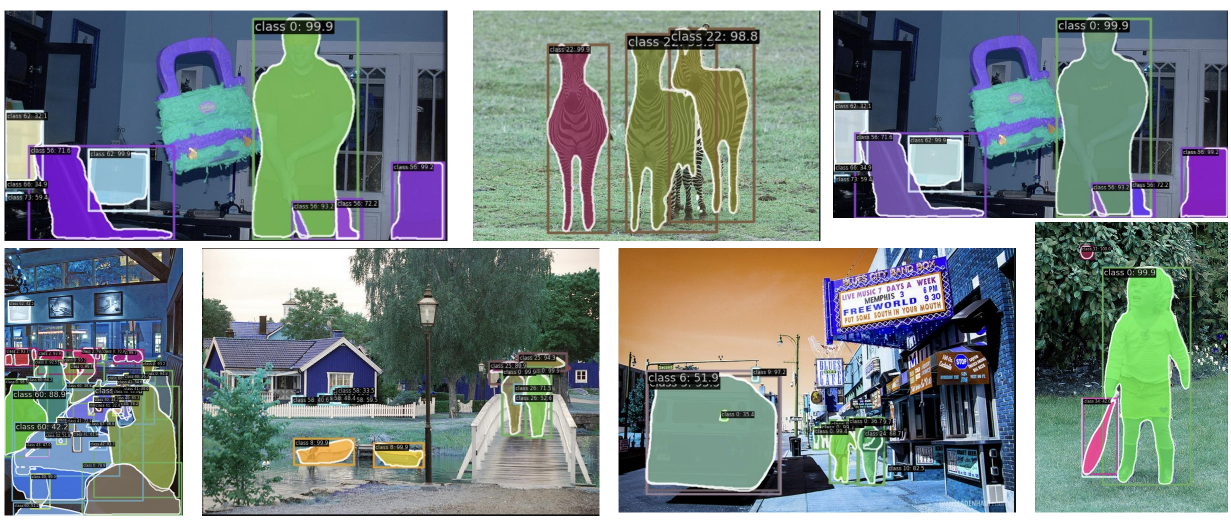

Figure 6: Qualitative results for object detection and instance segmentation on COCO-2017 using WCANet as the backbone framework.

enabling evaluation of both object detection and instance segmentation under a unified benchmark.

WCANet-T8 is first pre-trained on ImageNet-1K and then used as the backbone of RetinaNet [72] and Mask R-CNN [73]. The downstream models are trained for 12 epochs using the 1× schedule with a total batch size of 16. We employ AdamW [74] with an initial learning rate of 1×10^{-4} and a weight decay of 0.05. The learning rate is linearly warmed up during the first 500 iterations and is subsequently decayed using cosine annealing. Gradient clipping with a maximum norm of 0.1 is applied throughout training. No additional task-specific hyperparameter tuning is performed.

Table 4 shows the comparison between WCANet and representative models. WCANet consistently achieves higher average precision (AP) than conventional CNN baselines under comparable model complexity. Specifically, WCANet-T8 outperforms the standard ResNet50 [2] by +2.3 AP^b with RetinaNet, and by +2.0 AP^b and +2.3 AP^m with Mask R-CNN. Moreover, WCANet-T8 shows competitive performance against the efficient CNN architecture FasterNet-S [27], achieving +0.1 higher AP^b (40.0 vs.39.9) but slightly lower AP^m (36.7 vs.36.9) under Mask R-CNN.

To provide a qualitative evaluation of the proposed network, we visualize the object detection and instance segmentation results in Fig. 6. Notably, WCANet exhibits robust detection and precise segmentation even for small and texture-rich objects. In LWSA, the HH sub-band is preserved and directly participates in inverse wavelet reconstruction, while additional convolutional processing on HH is omitted to reduce computational overhead. The visualization and quantitative results (e.g., 22.5% AP_S in RetinaNet) indicate that this selective processing strategy still preserves the high-frequency spatial information required for dense prediction, including fine edges and object boundaries. Therefore, WCANet maintains strong sensitivity to small objects and fine spatial structures while avoiding unnecessary computation on the HH branch.

4.4. Medical Image Segmentation on ClinicDB

Table 5

Performance comparison on the ClinicDB dataset over five independent data splits. For each split, 80% of the images are used for training, 10% for validation, and 10% for testing. For each data split, all compared models use the same training, validation, and testing subsets and the same model-training seed (2026). The results are reported as mean ± standard deviation across the five data splits.

Method	Val Dice (%)	Val IoU (%)	Test Dice (%)	Test IoU (%)
MK_UNet	89.91 ± 0.63	83.46 ± 0.93	93.59 ± 0.98	88.46 ± 1.43
WCANet (Fixed Haar Filters)	90.95 ± 0.39	84.91 ± 0.56	93.75 ± 0.67	88.77 ± 0.72
WCANet (Learnable Filters, Ours)	**91.16 ± 0.60**	**85.43 ± 0.87**	**94.28 ± 0.60**	**89.26 ± 0.75**

To evaluate the cross-domain adaptability of WCANet, we further conduct medical image segmentation experiments on the CVC-ClinicDB dataset [75]. ClinicDB contains 612 RGB colonoscopy images with corresponding pixel-level polyp segmentation masks. All images have an original resolution of 384×288, and the task is formulated as binary polyp-versus-background segmentation. Compared with natural-image benchmarks, the dataset exhibits substantially different texture, contrast, illumination, specular reflections, and boundary characteristics. For training, all images are resized to 352 × 352, while the complete preprocessing, augmentation, optimization, and loss settings are provided in the Appendix.

Given the relatively small size of ClinicDB, results obtained from a single data partition may be sensitive to the particular training, validation, and testing split. We therefore conduct experiments over five independently generated data splits. For each split, the 612 images are divided into 489 training images, 61 validation images, and 62 test images, corresponding to approximately 80%, 10%, and 10%, respectively. For each data split, all compared models use exactly the same training, validation, and testing subsets and follow the same training protocol. The model-training seed is fixed to 2026 for all compared methods, and the results are reported as mean ± standard deviation across the five independent data splits.

As shown in Table 5, MK_UNet [76] achieves 93.59 ± 0.98% Test Dice and 88.46 ± 1.43% Test IoU. Introducing fixed Haar filters into WCANet yields 93.75 ± 0.67% Test Dice and 88.77 ± 0.72% Test IoU. In contrast, WCANet equipped with our proposed learnable wavelet filters achieves the best average performance, reaching 94.28 ± 0.60% Test Dice and 89.26 ± 0.75% Test IoU. Compared with the fixed-Haar variant, the learnable filters improve the average Test Dice and Test IoU by 0.53 and 0.49 percentage points, respectively. Meanwhile, the standard deviations of the learnable-filter variant remain comparable to those of the fixed-Haar counterpart across different data partitions, indicating that the observed performance improvement is robust to variations in the dataset split rather than being specific to a single partition.

These results further demonstrate the cross-domain adaptability of the proposed learnable wavelet filters. The learnable decomposition can adjust to the distinctive texture, contrast, and boundary characteristics of medical images. Meanwhile, the orthogonality constraint maintains a mathematically structured and non-redundant multi-band decomposition during optimization.

To intuitively analyze the effectiveness of our proposed WCANet in spatial feature extraction, we visualize the intermediate feature responses of MK-UNet, WCANet with fixed Haar filters, and WCANet with learnable filters in the first row of Fig. 7, together with representative segmentation results in the subsequent rows. Compared with MK-UNet, the wavelet-based variants suppress more irrelevant background responses, while the learnable-filter variant further produces a more compact and target-focused activation pattern. This observation suggests that adaptive frequency decomposition can better preserve task-relevant information and suppress less informative responses, which is consistent with the general feature-engineering principle of improving the discriminative quality of learned representations [77]. The segmentation examples further show that WCANet with learnable filters generally produces cleaner target regions and more accurate localization than the baseline and fixed-filter variants, supporting the effectiveness of the proposed Learnable Wavelet Spatial Attention for spatial feature modeling.

4.5. Model Analyses

In this subsection, we conduct controlled ablation experiments on ImageNet-100, a 100-category subset of ImageNet-1K. All ablation variants are trained for 100 epochs using exactly the same dataset partition, preprocessing, optimization, and evaluation protocol, such that only the architectural component under investigation is changed in each controlled comparison. Unless otherwise specified, WCANet-T4 is used as the default model for the ablation studies.

Analysis of Learnable Wavelet Filters. To comprehensively characterize both the individual contributions and cross-stage interactions of learnable wavelet filters, we conduct an exhaustive combinatorial ablation on WCANet-T4 covering all possible configurations across Stage 0, Stage 1, and Stage 2. As shown in Table 6, the fixed-filter baseline achieves 78.4% Top-1 accuracy. Enabling learnable filters at Stage 0, Stage 1, and Stage 2 individually improves the accuracy to 78.9%, 79.2%, and 79.4%, corresponding to gains of 0.5, 0.8, and 1.0 percentage points, respectively. The progressively larger improvements from Stage 0 to Stage 2 indicate that learnable wavelet filters become increasingly

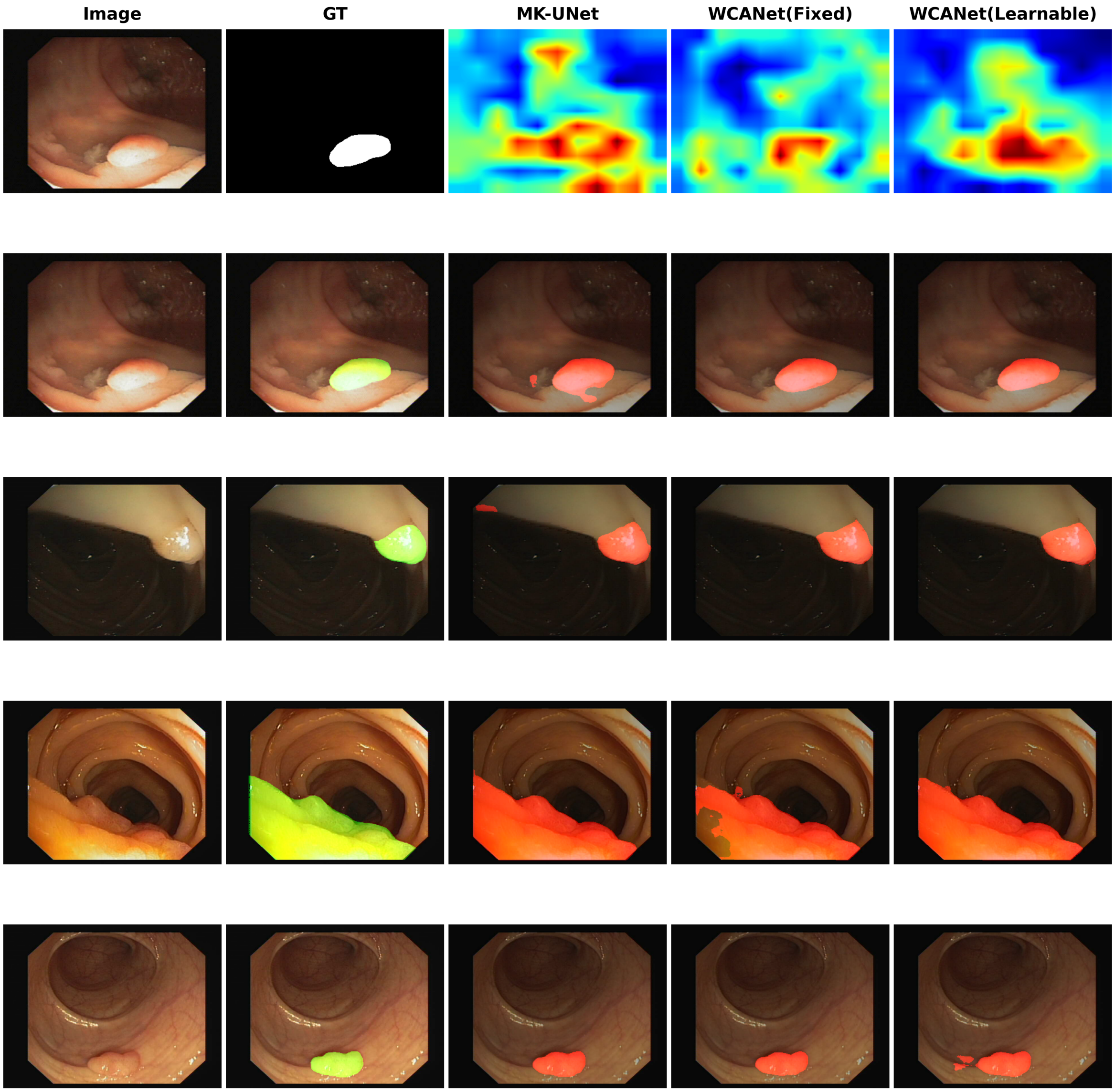

Figure 7: Qualitative feature-response and segmentation comparison on the ClinicDB dataset. The first row visualizes the intermediate spatial responses of MK-UNet, WCANet with fixed Haar filters, and WCANet with learnable filters, with the input image and ground truth provided as references. The following four rows present qualitative segmentation results. In the representative cases, the learnable-filter variant produces more target-focused feature responses and cleaner segmentation masks. The last row additionally presents a challenging example in which the improvement over the fixed-filter variant is less pronounced.

beneficial at deeper network stages, where feature representations are more abstract and semantically complex and therefore can benefit more from adaptive frequency decomposition. The pairwise configurations further reveal clear cumulative and complementary effects across stages. Specifically, Stage 0 & Stage 1 achieves 79.7%, Stage 0 & Stage 2 reaches 80.1%, and Stage 1 & Stage 2 obtains 80.4%, all outperforming their corresponding single-stage configurations. Among the pairwise settings, Stage 1 & Stage 2 provides the highest accuracy, indicating that jointly adapting the middle and deeper stages yields particularly effective cross-stage complementarity. Finally, enabling learnable filters at all three stages achieves the best Top-1 accuracy of 80.6%, improving the fixed-filter baseline by 2.2 percentage points. Taken together, these results show that the performance gains from learnable wavelet filters are broadly cumulative across network stages and exhibit complementary cross-stage interactions, while not following a strictly linear additive relationship.

Analysis of Wavelet Spatial Attention. To verify the rationality of the Wavelet Spatial Attention (WSA) design proposed in LWSA, we compare LWSA with different spatial feature extraction strategies. Specifically, we consider an identity transformation, 7×7 depthwise convolution, and the LS (Large-Small) convolution from LSNet [66] as alternative designs. As shown in Table 7, LWSA achieves the highest Top-1 accuracy of 80.6% with only 313M FLOPs, outperforming both the 7×7 depthwise convolution and

Table 6
Combinatorial ablation study of learnable filters across different stages on WCANet-T4.

Configuration	FLOPs (M)	Top-1 Acc (%)	Δ (%)
Baseline (None)	313.5	78.4	–
Stage 0 Only	313.5	78.9	+0.5
Stage 1 Only	313.5	79.2	+0.8
Stage 2 Only	313.5	79.4	+1.0
Stage 0 & 1	313.5	79.7	+1.3
Stage 0 & 2	313.5	80.1	+1.7
Stage 1 & 2	313.5	80.4	+2.0
All Stages	313.5	80.6	+2.2

Table 7
Comparison of LWSA with different spatial feature extraction strategies on WCANet-T4.

Strategy	FLOPs	Top-1
Identity	313	75.2
7×7 DW	318	79.1
LS Conv	321	80.4
LWSA	313	**80.6**

LS Conv while requiring lower computational cost. These results demonstrate that the proposed WSA provides an effective accuracy–efficiency trade-off for spatial feature modeling. This advantage can be attributed to its frequency-aware design, which selectively enhances informative subbands instead of applying the same spatial operation uniformly to the entire feature map.

Table 8
Ablation study of HH sub-band processing on ImageNet-100 and ClinicDB. Top-1 accuracy is reported for ImageNet-100, while Test Dice is reported for ClinicDB. Results are reported as mean ± standard deviation over five independent experiments.

Dataset	HH Strategy	FLOPs (M)	Result (%)
ImageNet-100	HH DWConv	318	80.61 ± 0.35
	HH Identity	**316**	$\mathbf{80.63 \pm 0.29}$
ClinicDB	HH DWConv	357	94.19 ± 0.41
	HH Identity	**354**	$\mathbf{94.21 \pm 0.67}$

Analysis of HH Sub-band Processing. In LWSA, the HH sub-band is preserved through an identity path and directly participates in the inverse wavelet transform, while additional convolutional processing is omitted to reduce computational overhead. To isolate the effect of HH processing, we construct a controlled comparison in which LL, LH, and HL are processed identically in both configurations, while only the treatment of the HH sub-band is changed. Given the importance of rigorous empirical validation, particularly in healthcare-related machine-learning applications [78], we evaluate this design on both ImageNet-100 and ClinicDB to examine its effectiveness in natural-image classification and medical-image segmentation. As shown in Table 8, on ImageNet-100, HH Identity reduces the FLOPs from 318M to 316M while achieving a comparable Top-1 accuracy of 80.63 ± 0.29

Table 9
All variants are based on WCANet-T4 and evaluated on ImageNet-100 with 100 training epochs.

Function	60 Epoch Top1-Acc(%)	100 Epoch Top1-Acc(%)
HL	73.01%	78.04%
HH	73.16%	77.92%
LL	71.92%	77.31%
$HL \odot LH$	77.58%	80.57%
$HH \odot LH$	76.71%	80.64%
$HL \odot HH$	76.64%	80.61%
$HL \odot LH \odot HH$	76.21%	80.54%

Analyze the effectiveness of different feature weight generation methods. We evaluated the impact on model performance when feature weights are generated from one, two, or three subbands. The experimental results are shown in Table 9. It can be observed that feature weights derived from a single subband yield the lowest Top-1 accuracy, whereas those generated from two or three subbands achieve better performance. However, compared to other approaches, the $HL \odot LH$ variant demonstrates superior convergence over variants using three or other combinations of subbands. This strongly validates the effectiveness of the $HL \odot LH$ method.

Table 10
Ablation study of individual components (ESE and LWSA) within the CCSA block.

Configuration	Top-1 Acc (%)
Baseline (None)	80.2
ESE Only	82.7
LWSA Only	83.3
CCSA (ESE + LWSA)	**85.5**

Ablation Study of CCSA Components. To isolate and quantify the individual contributions of the ESE module and the Learnable Wavelet Spatial Attention (LWSA), we conduct a component-wise ablation study on the CCSA block. As shown in Table 10, both components independently improve model performance. Compared to the baseline model (80.2%), introducing only the ESE module or only the LWSA module yields significant improvements of +2.5% and +3.1%, respectively. The slightly better independent performance of LWSA (83.3%) over ESE (82.7%) indicates that our wavelet-based spatial attention plays a particularly crucial role in visual feature extraction. Furthermore, combining ESE and LWSA into the complete cascaded CCSA block achieves the highest Top-1 accuracy (85.5%). This confirms that channel-wise and spatial-level feature enhancements are highly complementary, and the cascaded design is strictly necessary for comprehensive and effective feature extraction.

Table 11
Ablation study of filter types on COCO object detection (RetinaNet 1x schedule).

Filter Type	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
Fixed Haar Filters	37.8	58.2	40.2	21.3	40.9	51.0
Learnable Filters (Ours)	**38.6**	**59.0**	**41.1**	**22.5**	**41.5**	**51.7**
Degradation (Δ)	*-0.8*	*-0.8*	*-0.9*	*-1.2*	*-0.6*	*-0.7*

Impact of Learnable Filters on Downstream Tasks. To further validate the effectiveness of our learnable wavelet filters in dense prediction, we conducted an ablation study on the COCO dataset using the RetinaNet framework by replacing the learnable filters with fixed Haar filters. As shown in Table 11, reverting to fixed filters leads to a noticeable degradation in overall performance (AP drops by 0.8%). Notably, the accuracy degradation differs across object categories. The most severe drop occurs in small objects (AP_S decreases by 1.2%), followed by large objects (-0.7%) and medium objects (-0.6%). This scale-aware degradation strongly indicates that data-driven learnable filters are particularly crucial for adaptively distinguishing fine-grained details and subtle edges from background noise, an adaptability that rigid fixed filters fail to provide.

5. Conclusion

In this work, we propose WCANet, a lightweight neural network that achieves efficient feature modeling at a low computational cost. The core of WCANet is the Cascade Channel-Spatial Attention module (CCSA). CCSA decomposes feature extraction into the channel-wise and spatial levels. In the channel-wise, CCSA employs the ESE Module to augment key feature channels. In the spatial dimension, CCSA employs Learnable Wavelet Spatial Attention (LWSA) to achieve efficient and effective spatial modeling. CCSA enables WCANet to better focus on key features under low computational cost. WCANet achieves state-of-the-art speed-accuracy trade-offs on multiple devices.

A. Appendix

In this appendix, we provide more details on the experimental setup, architecture configuration, implementation of LWSA, limitations, and future work.

A.1. ImageNet-1K experimental settings

For the ImageNet-1K classification experiments, the detailed training configuration is summarized in Table 12. WCANet is trained from scratch without knowledge distillation or additional pre-training data. The training and evaluation resolutions are both set to 256 × 256. During training, images are processed using random resized cropping with an area range of (0.08, 1.0) and an aspect-ratio range of (3/4, 4/3), followed by random horizontal

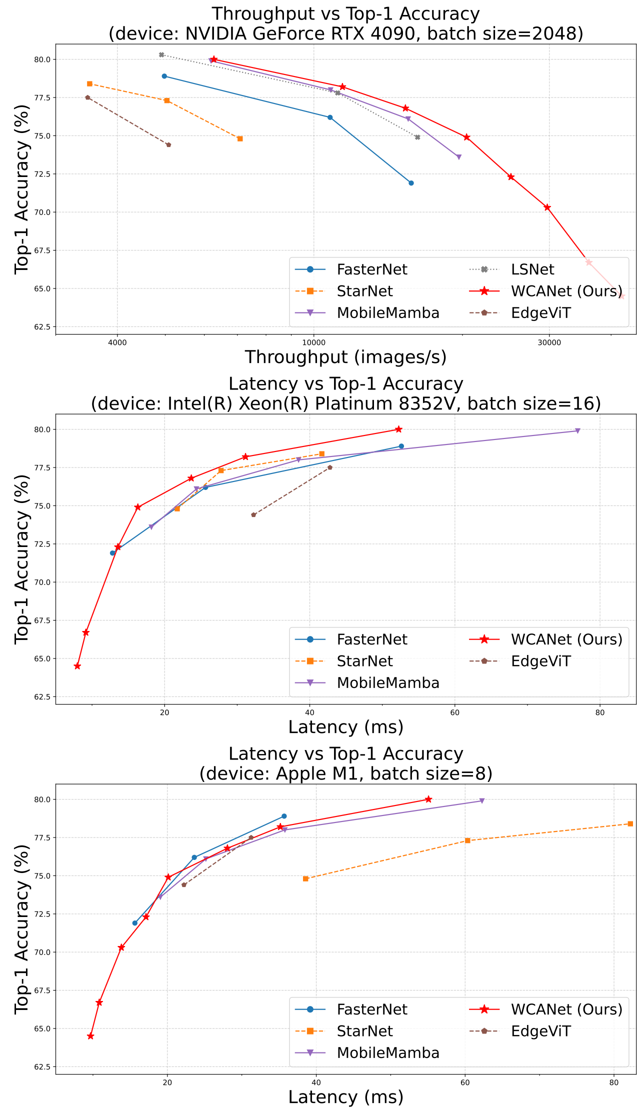

Figure 8: Accuracy–efficiency trade-offs of WCANet and representative models across different hardware platforms. GPU throughput is evaluated on an NVIDIA GeForce RTX 4090 with a batch size of 2048, while CPU and ARM latency are evaluated on an Intel(R) Xeon(R) Platinum 8352V with a batch size of 16 and an Apple M1 (ARM-based chip) with a batch size of 8, respectively. Each point is obtained from the mean of five repeated measurements, with the corresponding standard deviations reported in Table 3.

flipping with a probability of 0.5. RandAugment with the `rand-m9-mstd0.5-inc1` policy is then applied. The images are subsequently converted to tensors and normalized using the standard ImageNet mean $(0.485, 0.456, 0.406)$ and standard deviation $(0.229, 0.224, 0.225)$. Random erasing is further applied with a probability of 0.25 using pixel-level replacement. For evaluation, each image is first resized to $\lfloor 256/0.875 \rfloor = 292$, followed by a center crop of 256×256. The resulting image is then converted to a tensor and normalized using the same ImageNet mean and standard deviation as in training. We use AdamW as the optimizer

Table 12
Training Details on ImageNet

Variants	T1, T2, T2d5, T3, T4, T5	T6	T8
Train Res	256		
Test Res	256		
Epochs	300	400	
Batch size	1024		
Optimizer	AdamW		
Momentum	0.9/0.999		
LR	3e-3		
LR decay	cosine		
Warmup epochs	20		
Warmup schedule	linear		
Label Smooth	0.1		
Mixup	0.8		
Cutmix	1.0		
Random Erasing	0.25		
Dropout	✗		0.01
EMA	✓		

with an initial learning rate of 3×10^{-3} and a total batch size of 1024. A linear warmup is applied for the first 20 epochs, followed by cosine learning-rate decay. WCANet-T1, T2, T2d5, T3, T4, and T5 are trained for 300 epochs, while WCANet-T6 and WCANet-T8 are trained for 400 epochs because their training had not fully converged after 300 epochs. In addition to the above image-level augmentation, Mixup and CutMix are used with coefficients of 0.8 and 1.0, respectively, together with label smoothing of 0.1. An exponential moving average (EMA) of the model parameters is also employed during training.

A.2. ClinicDB Experimental Settings

For the ClinicDB experiments, all compared models follow the same training protocol. Input images are resized from the original 384×288 resolution to 352×352. Data augmentation includes random rotations of up to 90° and random horizontal and vertical flipping. Multi-scale training is performed using scales $\{0.75, 1.0, 1.25\}$. All models are optimized using AdamW with an initial learning rate of 5×10^{-4} and a weight decay of 1×10^{-4} for 200 epochs. A cosine annealing learning-rate schedule is adopted with a minimum learning rate of 1×10^{-6}, and the batch size is set to 8. Gradient clipping is applied with a maximum norm of 0.5. The training objective is the structure loss, consisting of weighted binary cross-entropy and weighted IoU loss. Dice and IoU are used as the evaluation metrics. For each of the five independent dataset splits, all compared models use the same training, validation, and testing subsets and the same training configuration.

A.3. Architectural Details

As shown in Tables 13 and 14, the stage-wise output stride of WCANet reaches ×64. Within the deepest-stage LWSA block, the wavelet transform further reduces the spatial resolution by a factor of 2, resulting in a minimum internal spatial resolution corresponding to an effective downsampling factor of ×128. To maintain exact spatial alignment during wavelet decomposition and inverse reconstruction, the input resolution should therefore be divisible by 128. Consequently, we adopt an input resolution of 256×256, avoiding additional padding and the associated computational and boundary effects.

Tab.13 presents the architectural details of the three-stage WCANet variants, and Tab.14 shows the details of the four-stage WCANet variants.

Table 13
Architectural details of WCANet-T1, WCANet-T3, and WCANet-T5.

Stage	Resolution	Type	Config	WCANet T1	WCANet T3	WCANet T5
stem	$\frac{H}{2} \times \frac{W}{2}$	Convolution	channels	8	13	16
	$\frac{H}{4} \times \frac{W}{4}$	Convolution	channels	16	26	32
	$\frac{H}{8} \times \frac{W}{8}$	Convolution	channels	32	52	64
	$\frac{H}{8} \times \frac{W}{8}$	Convolution	channels	64	104	128
1	$\frac{H}{16} \times \frac{W}{16}$	CCSA Block	channels	64	104	128
			blocks	2	2	3
2	$\frac{H}{32} \times \frac{W}{32}$	CCSA Block	channels	128	208	256
			blocks	3	3	4
3	$\frac{H}{64} \times \frac{W}{64}$	CCSA Block	channels	256	416	512
			blocks	2	2	3

Table 14
Architectural details of WCANet-T8.

Stage	Resolution	Type	Config	WCANet-T8
stem	$\frac{H}{2} \times \frac{W}{2}$	Convolution	channels	24
	$\frac{H}{4} \times \frac{W}{4}$	Convolution	channels	48
	$\frac{H}{8} \times \frac{W}{8}$	Convolution	channels	96
1	$\frac{H}{8} \times \frac{W}{8}$	CCSA Block	channels	96
			blocks	2
2	$\frac{H}{16} \times \frac{W}{16}$	CCSA Block	channels	192
			blocks	3
3	$\frac{H}{32} \times \frac{W}{32}$	CCSA Block	channels	384
			blocks	4
4	$\frac{H}{64} \times \frac{W}{64}$	CCSA Block	channels	640
			blocks	5

A.4. Limitation and Future Work

Limitation. Although WCANet achieves a favorable speed–accuracy trade-off across GPU, CPU, and ARM-based platforms, deployment on dedicated mobile and embedded accelerators remains to be further investigated. In addition, the current stage-dependent kernel-size schedule ($k = 7, 5, 3$) is fixed and may not optimally adapt to the spatial-frequency characteristics of every visual task. The current ClinicDB experiments provide initial evidence of the cross-domain adaptability of WCANet in medical image segmentation. Further evaluation on additional medical-image datasets will be considered in future work to assess its broader generalizability.

Future Work. We will validate the model's efficiency on edge devices and publish the results on GitHub. To further bridge the performance gap in downstream tasks, we will explore architectural fusion strategies that integrate WCANet with other paradigms without increasing the computational burden. Furthermore, we intend to investigate dynamic, multi-perspective feature learning strategies, drawing inspiration from recent advancements such as HMP-Former [79], to enhance the model's architectural flexibility and generalization capability.

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